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About

Dr. Zhou received her Ph.D. degree in Computer Science from Nanyang Technological University (NTU) in 2016. She received her MSE and BEng degrees from Beihang University in 2012 and 2009, respectively. She was a postdoc researcher in INRIA Rennes center (2016-2017) and a faculty member of Shenzhen University (2017-2023). Her research interests include parallel and distributed systems, cloud computing and high-performance computing. She has published more than 30 technical articles in refereed journals and conferences including SC, HPDC, ICS, ICDE, ICDCS, ICPP and TPDS. She has been actively serving the community by participating in the organizing/program committees for conferences including SC, HPDC, ICPP, Cluster and CIKM. She is also serving as an Associate Editor for IEEE Transactions on Parallel and Distributed Systems (TPDS) and an Editor for Future Generation Computer Systems (FGCS). She is the recipient of the IEEE-CS TCHPC Early Career Award and the ACM SIGHPC China Rising Star Award in 2021. She received the Shenzhen Young Scientist Award in 2023.

I am actively looking for PhD/RAs/Interns with strong background in high-performance computing (HPC) and system research. Please contact me with your CV.
**Strong preference will be given to candidates with hands-on experience in system-level development, performance optimizations, or participation in related competitions (e.g., ASC).*

Research Interests

- Parallel and Distributed Systems
- Cloud Computing
- High-Performance Computing

Selected Publications

- Sitian Chen, Amelie Chi Zhou*, Yucheng Shi, Yusen Li, Xin Yao, UpANNS: Enhancing Billion-Scale ANNS Efficiency with Real-World PIM Architecture. ***SC 2025***: 1-13
- Haobin Tan, Yao Xiao, Amelie Chi Zhou*, Kezhong Lu, Xuan Yang, Distributed and Adaptive Partitioning for Large Graphs in Geo-Distributed Data Centers. IEEE Trans. Parallel Distributed Syst. ***IEEE TPDS*** 36(6):1161-1174 (2025)
- Sitian Chen, Haobin Tan, Amelie Chi Zhou*, Yusen Li, Pavan Balaji, UpDLRM: Accelerating Personalized Recommendation using Real-World PIM Architecture. ***DAC 2024***: 1-6
- Amelie Chi Zhou, Zhoubin Ke, Jianming Lao, DyVer: Dynamic Version Handling for Array Databases. ***ICS 2023***: 144-154
- Amelie Chi Zhou, Juanyun Luo, Ruibo Qiu, Haobin Tan, Bingsheng He, Rui Mao, Adaptive Partitioning for Large-Scale Graph Analytics in Geo-Distributed Data Centers. ***ICDE 2022***: 2818-2830

- Amelie Chi Zhou, Ruibo Qiu, Thomas Lambert, Tristan Allard, Shadi Ibrahim, Amr El Abbadi, PGPregel: an end-to-end system for privacy-preserving graph processing in geo-distributed data centers. ***SoCC 2022***: 386-402
- Amelie Chi Zhou, Weilin Xue, Yao Xiao, Bingsheng He, Shadi Ibrahim, Reynold Cheng, Taming System Dynamics on Resource Optimization for Data Processing Workflows: A Probabilistic Approach. IEEE Trans. Parallel Distributed Syst. ***IEEE TPDS*** 33(1): 231-248 (2022)
- Amelie Chi Zhou, Jianming Lao, Zhoubin Ke, Yi Wang, Rui Mao, FarSpot: Optimizing Monetary Cost for HPC Applications in the Cloud Spot Market. IEEE Trans. Parallel Distributed Syst. ***IEEE TPDS*** 33(11): 2955-2967 (2022)
- Feng Zhang, Zheng Chen, Chenyang Zhang, Amelie Chi Zhou, Jidong Zhai, Xiaoyong Du, An Efficient Parallel Secure Machine Learning Framework on GPUs. IEEE Trans. Parallel Distributed Syst. ***IEEE TPDS*** 32(9): 2262-2276 (2021) (***Best Paper Award***)
- Amelie Chi Zhou, Yifan Gong, Bingsheng He, Jidong Zhai, Efficient process mapping in geo-distributed cloud data centers. ***SC 2017***: 16